Multiple Choice Questions – 4

C

What does the bit rate measure in digital communication?

- A) Number of symbols sent per second
- B) Number of signal elements per second
- C) Number of bits sent per second
- D) Frequency of the carrier wave

C

2. What is the relationship between bit rate N, symbol rate S, and bits per symbol r?

- A) N=S/r
- B) N=S+r
- C) N=S×r
- D) N=S−r

B

3. Which modulation technique alters the amplitude of the carrier signal?

- A) Frequency Shift Keying (FSK)
- B) Amplitude Shift Keying (ASK)
- C) Phase Shift Keying (PSK)
- D) Quadrature Amplitude Modulation (QAM)

D

4. In Quadrature Amplitude Modulation (QAM), what are the two main components of the signal?

- A) In-phase and out-of-phase components
- B) Amplitude and frequency components
- C) In-phase and quadrature-phase components
- D) Amplitude and phase components

A

5. Which of the following describes the Baud rate in digital communication?

- A) The number of signal elements per second
- B) The frequency of the signal
- C) The amplitude of the signal
- D) The modulation index

B

6. What is the symbol rate in Binary Phase Shift Keying (BPSK) if the bit rate is 10 Mbps?

OOK

$r=1$

$L=2$

- A) 5 Mbaud
- B) 10 Mbaud
- C) 15 Mbaud
- D) 20 Mbaud

Answer: B) 10 Mbaud

D

7. Which modulation scheme uses both phase and amplitude changes to represent data?

- A) Amplitude Shift Keying (ASK)
- B) Frequency Shift Keying (FSK)
- C) Phase Shift Keying (PSK)
- D) Quadrature Amplitude Modulation (QAM)

说明 Signal < noise

B

8. In a noisy communication channel with a very low signal-to-noise ratio, the channel capacity approaches:

$N=2B$

$C=B\log_2(1+SNR)$

$\frac{S}{N}$

- A) Infinity
- B) Zero
- C) Half the bandwidth
- D) Double the bandwidth

C

9. If the symbol rate is 2000 symbols per second and each symbol represents 3 bits, what is the bit rate?

$S=2000$

$r=3$

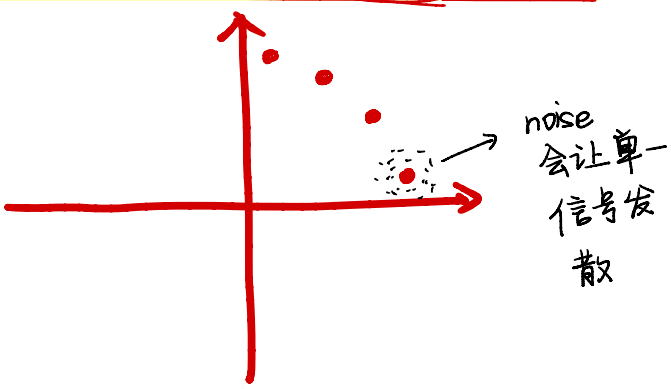
$N=S\cdot r=6000$

- A) 2000 bps
- B) 4000 bps
- C) 6000 bps
- D) 8000 bps

C

10. Which factor primarily limits the maximum number of symbols in M-PSK modulation?

- A) Signal frequency
- B) Amplitude range



- C) Susceptibility to noise
- D) Bit rate